

2.49 Find x in the diagram at left below. **Hints:** 355

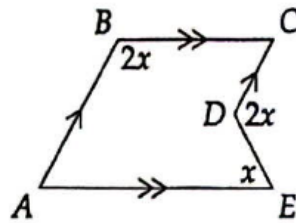


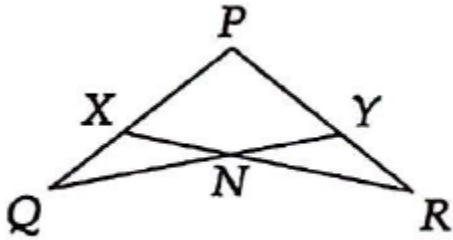
Figure 2.17: Diagram for Problem 2.49

2.54 It is possible for the interior angles of a triangle to be in the ratio $1 : 2 : 6$, but is it possible for the *exterior* angles of a triangle to be in the ratio $1 : 2 : 6$? Prove your answer. **Hints:** 14, 444

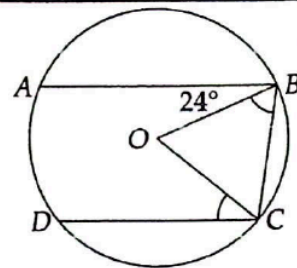
2.57★ Point Z is on side $\overline{PR}$ of $\triangle PQR$ such that $\angle PZQ = \angle PQZ$, and $\angle PQR - \angle PRQ = 42^\circ$. Find $\angle RQZ$.

3.4.4 In the figure, $PQ = PR$ and $\angle PQY = \angle PRX$.

(a) Prove that $QY = RX$.



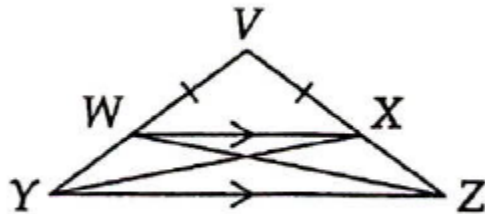
Problem 3.20: In the diagram, O is the center of the circle, $\overline{AB} \parallel \overline{CD}$, $\angle ABO = 24^\circ$, and $\angle OBC = \angle OCD$. Find $\angle BOC$.



3.6.6 Triangle ABC is equilateral in the diagram on the right above. Points D , E , and F are in triangle ABC such that $\angle CAD = \angle ABE = \angle BCF$. D lies on $\overline{CF}$, E lies on $\overline{AD}$, and F lies on $\overline{BE}$. Prove that triangle DEF is equilateral. **Hints:** 192, 574

3.6.7 In the diagram at right, $VW = VX$ and $\overline{WX} \parallel \overline{YZ}$.

(b) Prove that $YX = WZ$. **Hints:** 315



3.41 The measures of two angles of an isosceles triangle are $3x + 4^\circ$ and $x + 17^\circ$. Find all possible values of x . **Hints:** 148

3.45★ Let $ABCD$ be a square, and let E, F be points such that $DA = DE = DF = DC$ and $\angle ADE = \angle EDF = \angle FDC$. Prove that triangle BEF is equilateral. **Hints:** 280, 86, 575

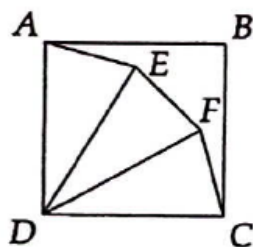
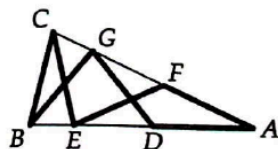
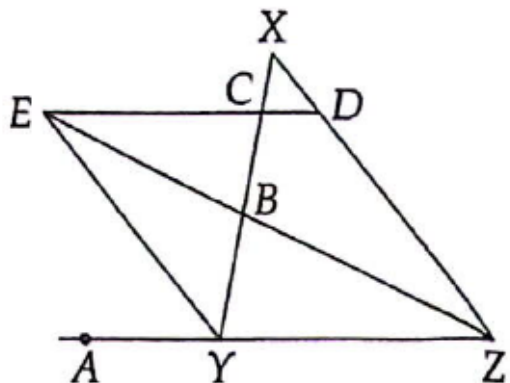


Figure 3.16: Diagram for Problem 3.45

3.46 In triangle ABC in the diagram below, D and E are points on side $\overline{AB}$, and F and G are points on side $\overline{AC}$, such that $AD = DG = GB = BC = CE = EF = FA$. Find $\angle BAC$. **Hints:** 436

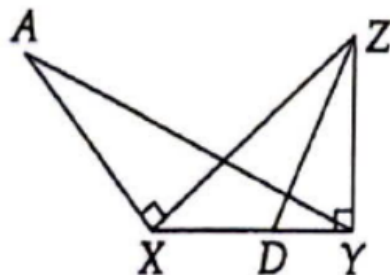


3.47 In the diagram at right, $\angle DZE = \angle EZY$, $\angle CYE = \angle AYE$, and $\overline{ED} \parallel \overline{YZ}$. Prove $CD = DZ - CY$. **Hints:** 399

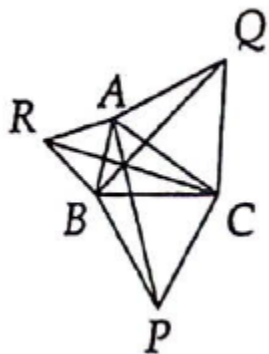


3.48 Describe under what conditions SSA Congruence 'works'. In other words, when is it true that if $AB = XY$ and $BC = YZ$ and $\angle BAC = \angle YXZ$, that we can immediately conclude that $\triangle ABC \cong \triangle XYZ$? (No proof is necessary; give a clear description of when SSA works and an intuitive explanation why it works in these cases.) **Hints:** 142

3.49 In the diagram at left below, $\angle XYZ = \angle AXZ = 90^\circ$, $AX = XY$, and $\angle XZD = \angle YZD$. Prove that $\overline{ZD} \perp \overline{AY}$. **Hints:** 197, 325



3.50 Draw equilateral triangles BCP , CAQ , and ABR outside $\triangle ABC$ as shown in the diagram at right above. Prove that $AP = BQ = CR$. Hints: 455



3.51★ There are two possible triangles with $AB = 13$, $BC = 10$, and $\angle A = 40^\circ$. What is the sum of the two possible values of $\angle B$? (Source: Mandelbrot) Hints: 9, 357, 163